

Rishi Tadavarthi

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EDUCATION

1227 S Dorsey Ln, Apt 101, Tempe, AZ, 85281

Arizona State University, Tempe, AZ

Spring 2023 - Fall 2024

Master of Science in Information Technology; GPA (4.33/4.00)

Anna University, Chennai, India

Fall 2016 - Fall 2020

Bachelors in computer science and engineering

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Java, JavaScript
- **Front-End:** HTML, CSS, ReactJS, AngularJS, Bootstrap, Flutter (DART), NodeJS
- **Back-End:** Django, Flask, Spring Boot
- **Cloud Platforms:** AWS, Google Cloud Platform, IBM, Heroku
- **Databases:** MySQL, NoSQL, MongoDB, PostgreSQL, RDS, Elastic Search
- **Machine Learning:** Logistic Regression, KNN, SVM, Random Forest, Neural Network, CNN, Recommendation systems, TensorFlow
- **Other Tools:** GitHub, Docker, Tableau, JIRA, ServiceNow

WORK EXPERIENCE

Assistant System's Engineer | Tata Consultancy Services LTD(TCS), Hyderabad, INDIA.

March 2021 – December 2022

- Developed different backend services (API's) using Python Programming language.
- Extensively used pandas, NumPy, SQL for data processing, analysis, and visualizations.
- Experience in developing web applications and implementing Model View Control (MVC) architecture using server-side applications Django and Flask.
- Skilled in manual and automated testing of applications and creating testing scenarios.
- Proficient in writing SQL queries using MySQL, MS SQL.
- Hands-on experience in executing MONGO DB general queries and aggregate queries.
- Develop products/features independently by writing scalable, efficient, and maintainable code.
- Extensive experience in Software Development Life Cycle (SDLC) and CI/CD pipelines.
- Actively participate in Sprint planning and execution

Software Engineer Intern | Tata Consultancy Services LTD(TCS), Hyderabad, INDIA.

May 2020 – July 2020

- Developed and implemented a machine learning and OCR-based solution to automate the extraction of handwritten text.
- Collaborated closely with a team of developers and data scientists to design and refine the project requirements.
- Conducted research and experimentation to identify and optimize machine learning algorithms for accurate text extraction.
- Gathered and preprocessed large datasets of handwritten text to train and fine-tune the OCR model.
- Worked on feature engineering and data augmentation techniques to enhance the model's performance and robustness.
- Evaluated the performance of the developed solution through rigorous testing and validation procedures, ensuring high accuracy and reliability.

Software Developer Intern & Software Developer |Kaashiv InfoTech, Chennai, India

July 2019 – December 2019

- I've developed Python-based backend services (APIs) using libraries such as pandas, NumPy, and SQL. These services are geared towards data processing, analysis, and visualization.
- I'm skilled in both manual and automated application testing, encompassing scenario creation. My proficiency extends to SQL and MongoDB for data retrieval.
- I've played a part in shaping efficient software development life cycle processes, including the implementation of Continuous Delivery, gaining approval from Audit, Compliance, and Risk teams. Also had hands-on experience in real-time data analytics and presenting insights through interactive dashboards.

Graduate Teaching Assistant(part-time) | Arizona State University, Tempe, AZ

August 2023 – present

- Working as Graduate Teaching Assistant for undergrad students in ASU for CSE-205(Object oriented programming and Data Structures) course.
- Assisting with professors Dr. Ryan Meuth and Dr. Lydia Fritz during lab sessions.
- Conducting student study hall sessions twice a week.

PROJECTS

Facial expression recognition

Fall 2022

- Developed Facial Emotion Recognition system using computer vision and deep learning techniques.
- Collected and curated diverse dataset of facial images representing different emotions.
- Performed data preprocessing, including resizing, normalization, and grayscale conversion.
- Utilized Haar-Cascade classifier for face detection and localization.
- Extracted meaningful features from facial images using techniques like facial landmarks, texture descriptors, and geometric ratios.
- Developed and trained a deep learning model based on Convolutional Neural Networks (CNN) for emotion classification, achieving reliable real-time emotion recognition.
- Skills: Software Deployment, CI/CD, Git, Computer Vision, Image Preprocessing, Facial Detection, OpenCV, CNN, HTML, Python

- Collected comprehensive dataset of Bangalore house listings, including area, amenities, and proximity to key facilities.
- Performed data preprocessing, cleaning, and feature engineering to ensure dataset quality.
- Utilized machine learning algorithms to train an accurate predictive model for house price estimation.
- Developed a user-friendly web interface using Flask, HTML, CSS, and JavaScript for intuitive user experience.
- Deployed the application on an AWS Ubuntu server, making it accessible through a web browser.
- Implemented CI/CD pipelines using GitHub Actions to automate software deployment and ensure code quality.
- Skills: Software Deployment, CI/CD, Git, Flask, HTML, CSS, JavaScript, AWS, GitHub, Python, Machine Learning

TexTrack**Fall 2023**

- Developed a file upload and text extraction web application using Flask, AWS S3, Textract, and MongoDB.
- Implemented functionality to upload files, extract text from documents using Textract, and store extracted text, along with user details, in a MongoDB database.
- Integrated AWS S3 for file storage and retrieval, allowing users to upload and download files from the application.
- Utilized AWS SQS to decouple the file upload process, enabling asynchronous processing and reducing user wait time.
- Implemented a message queue listener to retrieve messages from SQS, extract relevant information, and trigger the text extraction process.
- Implemented a secure authentication process using username and password credentials, ensuring that only authorized users can access the file upload and extraction functionalities.
- Ensured data integrity by checking for duplicate file names and avoiding overwriting existing files in the S3 bucket.
- Incorporated AWS SNS to send notifications about file uploads and extraction completion to subscribed users, enhancing user experience and engagement.
- Implemented a search functionality to retrieve and display user-specific files and their extracted text based on user-provided input.
- Implemented error handling and validation to handle various scenarios, such as missing file uploads, incomplete user information, and failed text extraction.
- Demonstrated proficiency in working with AWS SDK (boto3) to interact with AWS services, ensuring seamless integration and data flow between components.
- Successfully deployed the web application to a production environment, allowing users to securely upload files, extract text, and view extracted data.

HONORS/CERTIFICATIONS

- Recipient of the Ira A. Fulton Schools of Engineering Fellowship.
- Certified in Business English Vantage (council of Europe level B1) from Cambridge university.
- AWS Academy - Cloud Architecting, Cloud Security, Cloud Operations.
- Data Scientist certified in Datacamp.
- Python and Java Programmer certifications in NPTEL.
- Data science for everybody, Coursera.
- Applied Machine Learning in Python (University of Michigan), Coursera.